

Angle between two planes

Key Points

For two planes with normals n_1, n_2 the angle between them is the angle between the normals so

$$\cos \theta = \frac{n_1 \cdot n_2}{|n_1||n_2|}$$

Basic Skills

Q1. For the following planes state the normal

a) $2x - 3y + 5z = 10$

b) $2y + z = 10 + 2x$

c) $\left(r - \begin{pmatrix} 2 \\ 5 \\ 0 \end{pmatrix} \right) \cdot \begin{pmatrix} 2 \\ -5 \\ 1 \end{pmatrix} = 0$

d) The plane has the origin and points $(1,2,3)$ and $(-1,-5,2)$

e) Two in-planar lines: $r = \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix}$ and $r = \begin{pmatrix} 2 \\ 0 \\ 6 \end{pmatrix} + t \begin{pmatrix} 1 \\ 0 \\ 5 \end{pmatrix}$

Q2. Work out the angle between $a = \begin{pmatrix} 4 \\ 1 \\ -3 \end{pmatrix}$ and $b = \begin{pmatrix} 5 \\ -1 \\ 2 \end{pmatrix}$

Competency Skills

Q3. Plane Π_1 is $2x - 3y - z = 9$ and plane Π_2 is $x + 6y + 2z = 20$

a) Find the normals n_1, n_2

b) Hence find the angle between the planes

Q4. Find the angle between the planes $x + y + z = 4$ and $x + 2y - z = 5$

Q5. Two planes both contain the line $r = \begin{pmatrix} 5 \\ 1 \\ 3 \end{pmatrix} + t \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$ with Π_1 passing through the origin and Π_2 passing through $(0,1,1)$

a) Find the normals n_1, n_2

b) Hence find the angle between the planes

Advanced Skills

Q6. Explain if using different normals that describe the same plane affect the angle calculated between the planes

Q7. Two planes are given by $x + y + z = 0$ and $x + y - kz = 0$ for some constant k

a) Determine where the planes intersect for any k

b) Find the constant k such that acute angle between the planes is the largest possible

c) For $k = 1$ find the angle between the planes

d) For $k = 1$ find when the planes intersect the plane $z = 2$

e) Hence for $k = 1$ find the volume enclosed by the planes and $0 \leq x \leq 2, -4 \leq y \leq 2$ and $z = 2$

Extension

Q8. Show that the angle between lines in 2D is actually the angle between the normals of the lines